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MERCED PROPERTY AND CASUALTY COMPANY: HOW CLIMATE RISK LED TO ITS FAILURE

Yuan (Sabrina) Du, Olga Kanj, and Zhe Peng wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

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On September 29, 2021, Jessy Picado, chief executive officer (CEO) of Insuredhouse, a small property and casualty insurance company located in Lethbridge, Alberta, stepped into her office for the first time after the pandemic-era isolation. The first thing she did was check the AXA Future Risks Report, which highlighted the return of climate change to the top of the emerging risk list (see Exhibit 1).[1] Picado was aware that climate change and cybersecurity posed a threat to all industries, with the financial sector being at very high risk.[2] Because Picado had only recently assumed the CEO role, she had not yet had the chance to discuss climate risk mitigation with Dany Voisin, chief risk officer, as the company's focus for the past year had been on the pandemic and cybersecurity.

Picado had to review the company's strategies and prepare for the annual board meeting, and this included discussing with Voisin how climate risk would affect the company's operational goals, such as expansion to adjacent provinces. The province of Alberta, Insuredhouse's market, was exposed to the same risk that had plagued a California-based insurer, Merced Property and Casualty Insurance Company (Merced), in 2018. Analyzing the causal link between the risk—identified as the Camp Fire—and the failure of Merced was a good starting point from which to strengthen Insuredhouse's risk management. Picado assigned analysis of the Merced case study to Voisin's risk management team.

CAMP FIRE

For the previous 45 years, summers in California had been becoming longer, hotter, and drier. Since the 1970s, the average daytime temperature across the state had gone up by 1.4°C. For the mountainous Sierra Nevada region, the sultry summer weather continued into the fall season, causing 2.3 times more wildfires between October and December over 1996–2017 compared to 1972–1995.[3] North of the Sierra Nevada region was California's Central Valley, which comprised a few smaller cities and towns, including the state capital, Sacramento, and Butte County, only 80 miles north of Sacramento (see Exhibit 2).

At the heart of Butte County was the town of Paradise, known for its rich vegetation and hilly scenery. Home to 40,000 residents, Paradise hosted a vulnerable population: around 20 per cent of its residents were disabled and over a quarter were aged over 65.[4]

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The fall of 2018 in Paradise was particularly dry. From October 1 to November 7, the town received just 3 per cent of its average annual rainfall.[5] The last visible precipitation—more than 0.25 inch within a day—had fallen on April 7.[6] The low humidity due to a lack of rain led to parched woods that were perfect fuel for a wildfire.

On November 8, 2018, at around 6:30 a.m., an aged power line owned by California's largest utility firm, Pacific Gas and Electric Company (PG&E), ignited a fire. The incident happened around Camp Creek Road at Pulga, resulting in the name Camp Fire.[7] On windy days, PG&E could opt to shut down its power lines to contain the possible damage, but on that day, it had not done so.

In a few minutes, the fire had turned nearby woods to embers. Wind gusts at the time were reaching speeds of 50 miles per hour. These winds carried the embers further, which in turn ignited more trees, creating a firestorm.[8] The Camp Fire quickly engulfed Concow, Magalia, and Paradise, three neighbouring towns in Butte County, with Paradise hit the hardest.[9]

At 8:02 a.m., the California Department of Forestry and Fire Protection (Cal Fire) issued a mandatory evacuation order for Paradise. However, the alarm system at Butte County was not working, and many residents were unaware of the evacuation order. It was estimated that only 30 per cent of Paradise residents subscribed to the robocall emergency warning system named CodeRed. The system, with the spread of the fire, failed to reach 60 per cent of subscribers. To evacuate, residents had to rely on word of mouth, firefighters, and loudspeakers, and by closely watching the direction in which the flames were moving. Additionally, Paradise was divided into 14 zones due to its limited road capacity. Although residents could have evacuated zone by zone, authorities failed to turn main highways out of the city into one-way roads. This oversight created a major traffic jam, delaying the progress of the rescue efforts and evacuation.[10]

Owing to a combination of natural disasters and human error, the Camp Fire turned out to be one of the most disastrous in California. It burned over 150,000 acres, killed 86 people, and destroyed 18,793 buildings (see Exhibit 3).[11] The property damage specifically from the Camp Fire amounted to US$18 billion,[12] around half of which was insured.[13]

MERCED PROPERTY AND CASUALTY COMPANY

The Camp Fire also impacted Merced. A small property and casualty insurance company, Merced was based in Atwater, California. Established in March 1906 as Merced Mutual Insurance Co., it sold property, automobile, marine, and farming insurance policies. However, due to the deteriorating economic conditions in California's Central Valley, Merced's financial performance had been declining since 2006. Worried about its own viability, Merced had agreed to an acquisition proposed by the United Heritage Financial Group, Inc. (UHFG) in April 2013.[14]

Before acquiring Merced, UHFG owned three other insurers: United Heritage Life Insurance Company (UHLIC) (since 1934), United Heritage Property & Casualty Company (UHPC) (since 1908), and Sublimity Insurance Company (since 1896) (see Exhibit 4).[15]

The acquisition by UHFG helped Merced improve its financial performance. From 2012–2017, Merced's financial indicators suggested it was a viable business. Merced maintained a lower average loss ratio than the industry average for that six-year period, and its combined ratio was only 3 per cent higher than the industry average. At the end of 2017, with assets totalling $23 million, Merced achieved a significant 23.8 per cent growth in its direct premiums compared with the previous year (see Exhibit 5).[16]

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Nevertheless, Merced's lines of business in California relied too much on homeowner insurance policies. At the end of 2017, this line of business accounted for 65.9 per cent of the direct premiums written ($4,472,446 of a total $6,781,948 in direct premiums written) and 60.5 per cent of the policies (4,264 of 7,044 total policies).[17] Merced also had concentrated risk in areas classified by Cal Fire as high-severity zones.[18] The lack of business and geographical diversification, however, was not unique to Merced but common for small, regional insurance companies.[19]

MEASURES TO PROTECT HOMEOWNERS IN CALIFORNIA

Since 1988, California had imposed a 7 per cent cap on the annual rate increase in premiums insurers could charge.[20] If the insurer's rate increase exceeded the 7 per cent threshold, policyholders could initiate a review process. This measure made California the country's fifth cheapest state in terms of homeowner insurance premiums.

While Californian insurers could work around this cap by seeking approval for a rate increase of 6.9 per cent, the increasing risks they faced over the years rendered even this maximum rate adjustment inadequate. Despite a continued rate increase in property insurance, insurers still had to reject coverage in risky areas or deny renewal of an existing policy in such areas. In October 2020, statistics from California's Department of Insurance showed that from 2018–2019, non-renewals had increased by 31 per cent, and that high-risk areas had borne the brunt—rising by 61 per cent.[21] To protect policyholders, California's state government passed the Wildfire Safety and Recovery Act in 2019, which imposed a one-year moratorium on cancellation or non-renewal of homeowner insurance policies.[22]

In California, high-risk homeowners who were unable to obtain insurance from the market could purchase policies with basic coverage from the California FAIR Plan Association.[23] FAIR Plan was a private entity created in 1968 that served as an insurer of last resort, yet the premiums it charged were much higher than those by conventional insurers.[24] Due to increased wildfire risk, more and more Californian homeowners had to resort to FAIR Plan. In November 2019, FAIR Plan was ordered to provide comprehensive homeowners insurance, but refused to do so.[25]

THE FALL OF MERCED

As a result of the Camp Fire and the resultant insurance claims, Merced's liabilities were estimated to be $87 million, far exceeding its $23 million in assets. Seeing its apparent insolvency, the state regulator placed Merced into liquidation on November 30, 2018, and a liquidation order was issued on December 3. At that time, 7,436 policyholders were seeking payment of their claims or a refund of premiums.[26]

One potential reason for Merced's insolvency was that it had not purchased sufficient reinsurance protection. Merced had two reinsurance programs: an individual property risk program and a catastrophe program, which included six reinsurance treaties. Under the catastrophe program, Merced reinsured catastrophic losses up to $9.7 million for claims in excess of $0.3 million.[27] The two programs, while giving Merced a potential recovery level of $17.165 million, were not enough to cover the liabilities arising from claims.

Reinsurance is a risk management strategy used by insurance companies to transfer a portion of their risk to other insurance entities called reinsurers. In essence, it is the insurers' insurance. The primary goal of reinsurance is to spread the risk of large losses across multiple parties, reducing the financial impact on any single insurer. All property and casualty insurers relied on insuring their disruptive losses through reinsurance mechanisms. The volume of reinsurance varied, depending on factors such as company

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strategies, economic conditions, frequency and severity of catastrophic events, and regulatory changes. There was no universal reinsurance adequacy level: the proper level was to be made in relation to each company's maximum probable loss (see Exhibit 6).

On December 10, 2018, the California Insurance Guarantee Association (CIGA) stepped in to take over claim responsibilities from Merced. CIGA was obligated to pay claims up to $500,000 per policy in an insurance failure.[28] Brad Roeber, executive director of CIGA, realized that CIGA did not have the expertise to handle the liquidation, admitting that no one at CIGA "knows how to adjust claims on property."[29] To expedite the processing of claims, Roeber hired former Merced staff to help deal with the liquidation.[30]

By March 2019, CIGA had paid out $66 million in losses, loss adjustment expenses, premium refunds, and incurred administrative expenses. Moreover, CIGA estimated an additional loss and loss adjustment reserve of $21 million. Merced's liquidator—the Insurance Commissioner of the State of California—was able to recover $11.165 million and concluded that the reinsurance sources had been exhausted.[31]

ANALYSIS

Voisin had not expected Merced, a solvent insurance company, to go bankrupt overnight. He wanted to find out whether Merced had indeed been financially vulnerable, or if it could have survived and the liquidation decision had been too hasty. He turned to the data, including Merced's financials and ratios for 2016–2017 to run the analysis (see Exhibit 8 and Exhibit 9).

The first step was to compute Merced's key financial ratios. However, in this highly regulated sector, insurance companies adopted a different set of performance indicators compared to non-insurance firms. Insurance companies also report their financial information under statutory accounting principles (SAP) instead of generally accepted accounting principles (GAAP) as used by other US companies (see Exhibit 10).

Voisin had to evaluate Merced's performance and conduct a comparative analysis contrasting Merced's financial ratios with those of other similar-sized US insurers to determine whether Merced was financially healthy. What would be the most useful indicators in this analysis? Was the Camp Fire the root or proximate cause that led to Merced's downfall? What are the implications for insurers facing similar climate risks? Voisin had to make recommendations to Insuredhouse's chief risk officer and CEO for how to adapt the existing risk management plan to avoid experiencing a similar fate as Merced.

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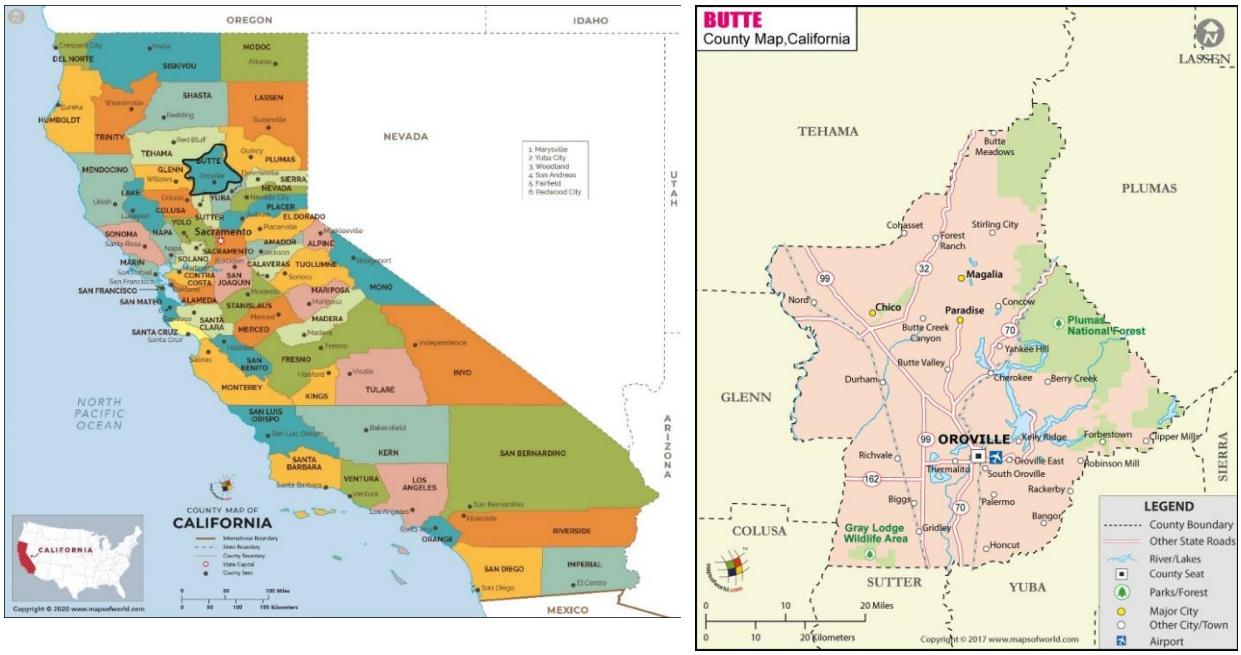
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EXHIBIT 1: TOP 10 EMERGING RISKS

	Top 10 emerging risks
1	Climate change
2	Cybersecurity
3	Pandemics and infectious diseases
4	Geopolitics
5	Social and local conflicts
6	Natural resources and biodiversity
7	New security and terrorism
8	Financial stability
9	Macroeconomics
10	Artificial intelligence and big data

Source: AXA, *AXA Future Risks Report 2021*, accessed December 28, 2023, https://www.axa.com/en/news/2021-future-risks-report.

EXHIBIT 2: BUTTE COUNTY, CALIFORNIA, USA

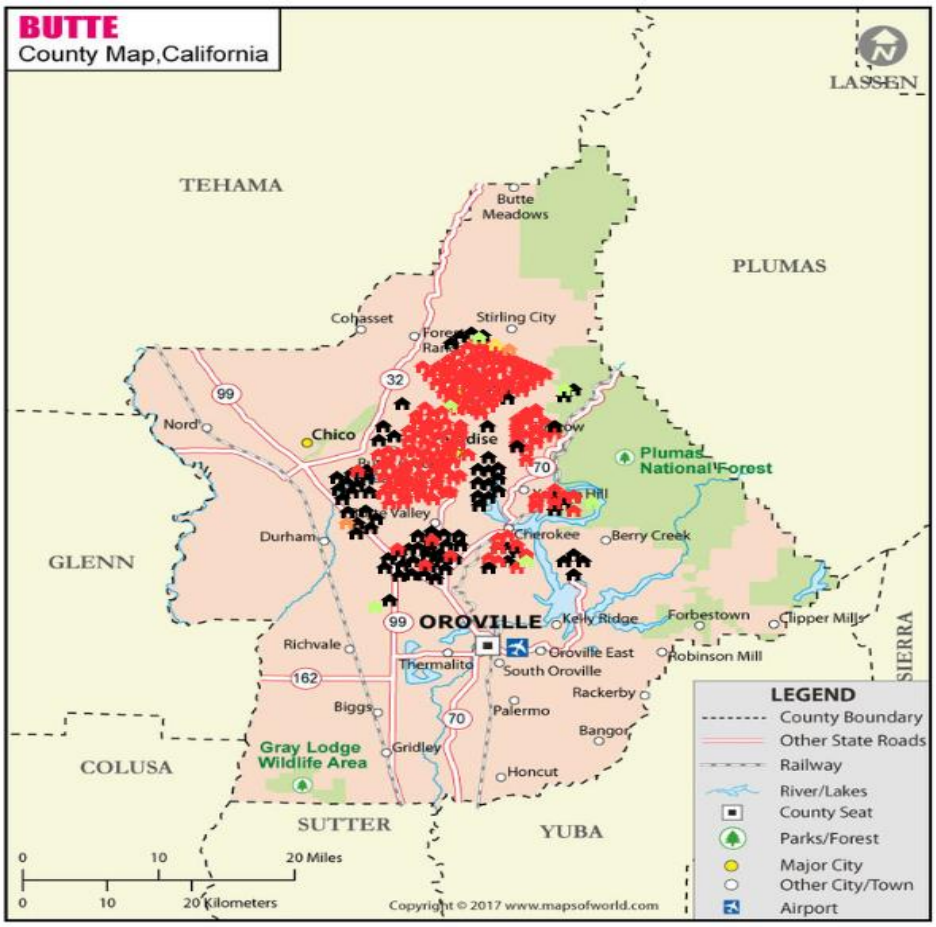


Notes: Pulga was located east of Concow; the towns most affected by the Camp Fire were Concow, Magalia, and Paradise.
Source: (Left) "County Map of California," accessed December 28, 2023, https://www.mapsofworld.com/usa/states/california/california-county-map.html. (Right) "Map of Butte County, California," assessed December 28, 2023, https://www.mapsofworld.com/usa/county-maps/california/butte-county-map.html. Used with permission.

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EXHIBIT 3: DAMAGE IN BUTTE COUNTY CAUSED BY THE 2018 CAMP FIRE



Notes: Pulga was located east of Concow; the towns most affected were Concow, Magalia, and Paradise; the red line showed the boundary of Butte County; the red shape showed the destroyed structures; the orange and yellow shapes marked structures with major and minor damage; the green shape showed slightly affected structures; and the black shape marked structures with no visible damage.

Sources: "Cal Fire Structure Status," Butte County, California, accessed December 28, 2023, https://buttecountyrecovers.org/cal-fire-structure-status; "Map of Butte County, California," assessed December 28, 2023, https://www.mapsofworld.com/usa/county-maps/california/butte-county-map.

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EXHIBIT 4: INSURERS IN THE UNITED HERITAGE FINANCIAL GROUP

Insurer	Founded in	In the group since	Type of insurer	Headquarters
United Heritage Life Insurance Company	1934	1934	Life and health	Meridian, Idaho
United Heritage Property & Casualty Company	1908	1908	Property and casualty	Meridian, Idaho
Sublimity Insurance Company	1896	2003	Property and casualty	Sublimity, Oregon
Merced Property and Casualty Company	1906	2013	Property and casualty	Atwater, California

Source: Created by the case authors based on SNL Database (2017).

EXHIBIT 5: MERCED PROPERTY AND CASUALTY—FACT SHEET

Panel A: Line of Business Breakdown (Direct Premiums Written)

	2017 ($)	2016 ($)
Homeowners	4,472,446	3,997,044
Auto liability and damage	1,609,470	796,434
Dwelling fire	584,940	565,299
Dwelling liability	115,092	117,910
Total	6,781,948	5,1476,687

Panel B: Number of Policy Contracts

	2017	2016
Homeowners	4,264	3,961
Auto liability and damage	1,137	581
Dwelling fire	935	941
Dwelling liability	708	724
Total	7,044	6,207

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EXHIBIT 5 (CONTINUED)

Panel C: Key Ratios—Industry Average vs. Merced

	Industry average			Merced
Key ratios	**2017**	**2016**	**2012–2017**	**2012–2017**
Loss ratio (SAP)	50.91	47.52	68.56	54.74
Expense ratio (GAAP)	35.54	35.57	40.10	56.66
Combined ratio	103.39	101.62	108.66	111.39
Investment ratio	9.42	10.14	9.08	16.16
Operating ratio	93.98	91.48	113.09	95.23

Notes: SAP = statutory accounting principles; GAAP = generally accepted accounting principles; expense ratio (GAAP) = other underwriting losses / net premium earned; loss ratio = (loss incurred + loss adjustment expenses incurred) / net premium earned; combined ratio = loss ratio + expense ratio; operating ratio = combined ratio – net investment ratio; investment ratio (SAP) = net investment income / net premium written.
Source: Created by the case authors based on SNL Insurance.

EXHIBIT 6: INDICATIVE MEASURES OF REINSURANCE ADEQUACY FOR US INSURERS COMPARABLE TO MERCED (2012–2018)

Comparable insurance companies were selected that had a similar size to Merced as measured by the premium written and the size of assets. Plymouth Rock Assurance Preferred Corporation and ARECA Insurance Exchange were two equally small insurers in terms of operating income, while United Heritage Property & Casualty Company and Sublimity Insurance Company shared the same parent company with Merced.

Panel A: Reinsurance Recoverable / Capital and Surplus (C&S) (%)

	2017	**2016**	**2015**	**2014**	**2013**	**2012**
Plymouth Rock Assurance Preferred Corporation	31.43	10.66	0.00	0.00	13.76	11.93
ARECA Insurance Exchange	6.39	9.21	12.51	10.00	8.96	15.84
United Heritage Property & Casualty Company	20.39	25.06	21.41	7.67	0.00	0.00
Sublimity Insurance Company	31.43	10.66	0.00	0.00	13.76	11.93

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EXHIBIT 6 (CONTINUED)

Panel B: Ceded Premium / Direct Premium Written (%)

	2017	2016	2015	2014	2013	2012
Plymouth Rock Assurance Preferred Corporation	9.34	9.20	9.19	8.90	8.84	7.38
ARECA Insurance Exchange	40.94	41.98	43.58	41.16	41.89	43.03
United Heritage Property & Casualty Company	26.28	30.81	16.35	16.10	16.11	17.96
Sublimity Insurance Company	22.14	13.92	10.27	9.39	11.84	10.42

Source: Created by the case authors based on SNL Insurance.

EXHIBIT 7: A.M. BEST'S RATING CHANGES IN UNITED HERITAGE PROPERTY & CASUALTY COMPANY AND ITS SUBSIDIARIES

Event	Date of rating change	Firm	Outlook	Financial strength rating	Long-term issuer credit rating	Change
	Before May 20, 2014	UHPC and three affiliates		B++ (Good)	BBB	
Merced acquired by UHPC	May 20, 2014	UHPC and three affiliates		A− (Excellent)	A−	Upgrade
Merced liquidated due to Camp Fire on November 8, 2018	December 3, 2018	UHPC	Negative	B+ (Good)	BBB–	Downgrade
		Merced		F (Liquidation)	F	Downgrade
		Sublimity Insurance Company	Negative	B++ (Good)	BBB	Downgrade
		United Heritage Life Insurance Company	Negative	B++ (Good)	BBB+	Downgrade

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EXHIBIT 7 (CONTINUED)

Event	Date of rating change	Firm	Outlook	Financial strength rating	Long-term issuer credit rating	Change
Update	December 6, 2019	UHPC	Stable	B+ (Good)	BBB–	Upgrade
		Sublimity Insurance Company	Stable	B++ (Good)	BBB	Upgrade
		United Heritage Life Insurance Company	Stable	B++ (Good)	BBB+	Upgrade
Update	February 5, 2021	UHPC	Positive	B+ (Good)	BBB–	Upgrade
		Sublimity Insurance Company	Positive	B+ (Good)	BBB	No change
		United Heritage Life Insurance Company	Positive	B++ (Good)	BBB+	Upgrade

Notes: UHPC = United Heritage Property & Casualty Company.
Source: A.M. Best Rating Services (2022).

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EXHIBIT 8: FINANCIAL STATEMENTS FOR MERCED (2016–2017)

Panel A: Statutory Statements of Admitted Assets, Liabilities, and Capital and Surplus (Balance Sheet) (in thousand $)

	2017	2016
Assets		
Investments:		
Bonds	19,669	18,879
Preferred stocks	320	311
Common stocks	945	1,056
Real estate	708	698
Cash and short-term investments	349	955
Total cash and investments	21,992	21,899
Premiums and considerations due	773	569
All other admitted assets	667	695
Total assets	**23,431**	**23,163**
Liabilities		
Loss reserves	1,312	935
LAE reserves	546	613
Total loss and LAE reserves	1,858	1,548
Unearned premium reserve	3,591	3,013
Commissions, other expenses, and taxes due	547	289
Derivatives	0	0
Payable to parent, subsidiaries, or affiliates	59	29
All other liabilities	545	758
Total liabilities	**6,600**	**5,637**
Policyholders' surplus		
Common capital stock	3,163	3,028
Surplus notes	1,607	2,712
Unassigned surplus	5,031	5,382
Other including gross contributed	7,031	6,404
Total policyholder surplus	**16,832**	**17,526**
Total liabilities and C&S	**23,431**	**23,163**

Notes: LAE = loss adjustment expenses; C&S = capital and surplus.

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EXHIBIT 8 (CONTINUED)

Panel B: Statutory Statements of Operations (Income Statement) (in thousand $)

	2017	2016
Underwriting revenue		
Personal P&C direct premiums	6,013	4,724
Commercial P&C direct premiums	769	753
Direct premium written	6,782	5,477
Net reinsurance premiums	(709)	(574)
Net premium written	6,073	4,903
Change in U/E premiums reserve	578	415
Net premium earned	**5,495**	**4,488**
Underwriting deductions		
Net losses paid—personal	1,723	814
Net losses paid—commercial	321	73
Total net losses paid	**2,044**	**886**
Net LAE paid	**1,009**	**1,047**
Change in loss reserves—personal	310	46
Change in loss reserves—commercial	67	134
Changes in LAE reserves	(67)	(29)
Net change in loss and LAE reserves	**310**	**151**
Losses and LAE incurred	3,363	2,085
Other underwriting expense incurred	3,234	2,877
Net underwriting gain (loss)	**(1,102)**	**(474)**
Investment income		
Net investment income	748	813
Net realized capital gains (losses)	20	116
Other income	69	34
Finance service charges	71	42
All other income	(2)	(8)
Total investment income	**837**	**963**
Net income		
Net income after capital gains (loss) before tax	(265)	489
Federal income tax	(85)	(79)
Net income	**(180)**	**568**

Notes: P&C = property and casualty; U/E = unearned; LAE = loss adjustment expenses.

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EXHIBIT 8 (CONTINUED)

Panel C: Statutory Statements of Changes in Capital and Surplus (Statement of Changes in Shareholders' Equity)

	Common stock		Paid-in and contributed surplus ($)	Certificate of contribution ($)	Unassigned surplus ($)	Total capital and surplus ($)
	Shares ($)	Value ($)				
Balance on January 1, 2016	1,101	3,027,750	6,404,322	2,819,420	4,621,779	16,873,271
Net income	–	–	–	–	567,210	567,210
Payments on certificates of contribution				(107,473)	–	(107,473)
Change in net unrealized capital gains and losses					97,846	97,846
Change in net deferred income tax					(467)	(467)
Change in non-admitted assets					95,890	95,890
Balance on December 31, 2016	1,101	3,027,750	6,404,322	2,711,947	5,382,258	17,526,277
Net loss	–	–	–	–	(179,989)	(179,989)
Redemption of certificates of contribution	49	134,750	626,530	(761,280)	–	–
Payments on certificates of contribution	–	–	–	(343,413)	–	(343,413)
Dividends to stockholder	–	–	–	–	(100,000)	(100,000)
Change in net unrealized capital gains and losses	–	–	–	–	49,361	49,361
Change in net deferred income tax	–	–	–	–	46,048	46,048
Change in non-admitted assets					(166,740)	(166,740)
Balance on December 31, 2017	1,150	3,162,500	7,030,852	1,607,254	5,030,938	16,831,544

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EXHIBIT 8 (CONTINUED)

Panel D: Statutory Statements of Cash Flow (Cash Flow Statement)

	2017 ($)	2016 ($)
Cash flow from operating activities		
Underwriting activities		
Premiums collected, net of reinsurance	5,872,499	4,718,084
Loss and loss adjustment expenses paid	(3,160,295)	(2,115,439)
Underwriting expenses paid	(3,250,830)	(2,852,075)
Net cash used in underwriting activities	(538,626)	(249,430)
Investment income received, net	594,362	674,515
Income taxes recovered	140,001	21,288
Other income	69,044	33,635
Net cash provided by operating activities	**264,781**	**480,008**
Cash flow from investing activities		
Payments for purchase of bonds	(2,247,918)	(3,477,691)
Payments for purchase of equity investments	(588,044)	(306,171)
Proceeds from sale or maturity of bonds	1,720,197	1,851,823
Proceeds from sale of equity investments	690,308	783,312
Payments for improvements of real estate	(28,635)	(42,535)
Payable for securities	–	39,546
Net cash used in investing activities	**(454,092)**	**(1,151,716)**
Cash flow from financing activities		
Principal payments for certificates of contribution	(343,413)	(107,473)
Redemption of common stock	–	(2,245)
Dividends to stockholder	(100,000)	–
Other cash provided	27,600	83,970
Net cash used in financing activities	**(415,813)**	**(25,748)**
Net decrease in cash and short-term investments	(605,124)	(697,456)
Cash and short-term investments at beginning of year	954,593	1,652,049
Cash and short-term investments at end of year	349,469	954,593

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EXHIBIT 8 (CONTINUED)

Panel E: Additional Notes

1. Direct Premium Written, Assumed Premium Written, and Gross Premium Written

	2017 ($)	2016 ($)
Direct premium written	6,781,948	5,476,687
Assumed premium written	288,890	280,383
Gross premium written	7,070,838	5,757,070
Ceded premium written	(997,709)	(853,919)
Net premium written	6,073,129	4,903,151
Direct premium written, previous year	5,477,687	4,833,000
Net premium written, previous year	4,903,151	4,389,000

2. Reinsurance Recoverable (Not the Balance Sheet Item)

	2017 ($)	2016 ($)
Reinsurance recoverable (flow)	747,831	887,765

Notes: Gross premium written = direct premium written + assumed premium written; assumed premium written during an accounting period is the part of premiums written by the insurer when acting as a reinsurer; net premium written = gross premium written – ceded premium written = direct premium written + assumed premium written – ceded premium written.
Source: Merced Annual Report (2017).

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EXHIBIT 9: KEY FINANCIAL RATIOS FOR INSURANCE COMPANIES

The ratios that are commonly used as profitability measures for insurance business

Ratio	Definition
Loss ratio	This measures the percentage of premium that has been used to pay for losses It is calculated as 1. Under SAP: (loss + loss adjustment expenses incurred) / net premium written 2. Under GAAP: (loss + loss adjustment expenses incurred) / net premium earned
Expense ratio	This measures an insurer's incurred underwriting expenses associated with the net premium It is calculated as 1. Under SAP: underwriting expenses / net premium written 2. Under GAAP: underwriting expenses / net premium earned
Combined ratio	This measures an insurer's overall underwriting profit It is calculated as loss ratio + expense ratio (typically using the measures under SAP)
Investment ratio	This measures the investment return for insurers It is calculated as 1. Under SAP: net investment income / net premium written 2. Under GAAP: net investment income / net premium earned
Operating ratio	This measures the overall operating profit after netting out the investment income It is calculated as: combined ratio – investment ratio

Notes: SAP = statutory accounting principles; GAAP = generally accepted accounting principles.
Source: Created by the author based on Insurance Regulatory Information System (iris) Ratios Manual, NAIC (2023).

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EXHIBIT 10: A COMPARISON OF SAP AND GAAP

	GAAP	SAP
Purpose	GAAP is a set of accounting standards used across various industries and sectors to prepare financial statements for external reporting. It is designed to provide consistency and comparability in financial reporting.	SAP is primarily used in the insurance industry to regulate and standardize the financial reporting of insurance companies. Designed to ensure the financial stability and solvency of insurance companies, it is often more conservative than GAAP.
Valuation of assets and liabilities	GAAP focuses on a more market-oriented approach to asset and liability valuation, which may result in more market-based values.	SAP places strong emphasis on the conservative valuation of assets and liabilities. Only highly liquid assets—known as admitted assets—can be included on the balance sheet. Other non-admitted assets such as furniture and fixtures, office machines, and other illiquid assets cannot be included as assets on the balance sheet.
Income and expense recognition	Under GAAP, expenses and premiums are recognized based on the matching principle. In other words, expenses should be recognized in the same coverage period as the net premium earned, which is treated as revenue.	Under SAP, expenses such as policy acquisition costs are recognized only when they are incurred, whereas premiums are earned over the life of the policy.

Notes: SAP = statutory accounting principles; GAAP = generally accepted accounting principles.
Source: NAIC Statutory Accounting Principles; “Key Differences between SAP and US GAAP," PWC, January 31, 2021, https://viewpoint.pwc.com/dt/us/en/pwc/accounting_guides/insurance-contracts/Insurance-Contracts/Ch13_statutory_accounting/13_5keydifferencesbetween.html.

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